CubeSat missions and applications

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Chantal Cappelletti and Daniel Robson
Department of Mechanical, Materials and Manufacturing Engineering, University of Nottingham, University Park, Nottingham, United Kingdom

1 Introduction

The key driving factors behind the worldwide adoption and growth of CubeSats can be approximately summarized as the following:

1. **Accessibility**
   Their original intention was to be an educational tool for university student teams; with this platform universities were able to afford the financial costs, development time, and expertise to design, launch, and operate a satellite over the course of a student's degree program. These simplicity and cost-effectiveness factors were achieved by creating a simplified design, using recommended affordable COTS components and accepted specifications and requirements that streamlined various stages of the development cycle such as deployment, structural design, and some verification requirements.
2. **Standardization**
   The incidental industry standards came about over time, thanks to CubeSats design specifications such as structural dimensions, deployment mechanisms, and the "stackability" of units. Within the boundary conditions of the design specification, a few engineering solutions became standout options and very quickly became industry standards—such as the PC/104 form factor for electronics and the P-POD common deployment mechanism. These enabled the further streamlining of the development process and mass production of parts and even entire subsystems as COTS products for CubeSats, further reducing costs to developers.
3. **Entrepreneurship**
   Over time the result of these net forces led to CubeSats becoming high return-on-investment space platforms, with low start-up or "buy-in" costs (costs referring to monetary costs and resources like time, expertise, equipment, and facilities). This is what enables the CubeSat today to fulfill such a wide variety of roles in such a wide variety of mission scenarios, from a wide variety of developers and customers.
4. **Technology**
   Advances in technology have allowed for more miniaturized, distributed autonomous, and higher-performing systems to be flown and more of this flight hardware to become COTS available products. Many areas of this book will detail some of these advances in materials, manufacturing, power, software algorithms, mission architectures (both in space and ground segments), communications, and more besides.
5. **Community**
   This spread of experience, technology, facilities, and industry has helped create a boom in services and customers and one that may not have evolved from traditional aerospace industries or practices. Opportunities for customers from less wealthy or less space-invested countries, industries, or research fields have broadened the scope of experiments and

CubeSat Handbook. https://doi.org/10.1016/B978-0-12-817884-3.00002-3

objectives that can be accomplished in space using a CubeSat. A plethora of smaller, more novel, and enterprising companies have formed to fill this niche to support such customers interested in taking their payloads to orbit, regardless of the stage of development—from feasibility consultations to inexpensive launch services. In business terms, this is referred to as a horizontal market (as opposed to "traditional" aerospace vertical markets and industries). This has led CubeSats to sometimes being referred to as a part of NewSpace, Space 4.0, or the "democratization" of Space.

With these key factors driving the spread of CubeSat technology across the world and beyond, we shall explore what was once regarded as an educational "toy" became a dominant force in the NewSpace industry and what roles it may play in the future [1–3].

2 Applications

The CubeSat specification by its engineered design and increasingly commonplace use in the space sector has led to CubeSats becoming highly versatile platforms. They can be used to achieve many different mission objectives inexpensively due to their low cost and simple development process and integration into mission architecture. Importantly, their lightweight aspect means they can be assigned as secondary payloads to many different orbits and destinations, and thanks to standardizations such as the P-POD deployment mechanism, with relative ease for the developer and launch service provider [4,5].

Further to this, services, technology, and facilities developed internationally to support this industry—COTS components, CubeSat-specific consultants, and other services—are now enabling opportunities for missions and customers that may have faced difficulty and high costs in developing payloads for traditional spacecraft or platforms or even accessing the space sector at all [1]. To highlight and summarize some of the areas in which CubeSats operate or are under development, the nonexhaustive list in the succeeding text has been compiled:

- **As alternative platforms for space-borne instruments**
  Enhancing traditional satellite roles such as telecommunications, observations of the Earth, and astronomical targets such as the Sun, galaxies, exoplanets, and many more scientific targets.
- **As alternative platforms for space-borne experiments and laboratories**
  Giving life sciences, pharmacology, materials science, radiology, and other fields access to space as an environment.
- **As affordable technology-demonstrator and proof-of-concept platforms**
  Due to their low costs to develop and the possibility to use existing components, CubeSats have found a natural application as in-orbit demonstrators for several kinds of new technologies (solar cells, radio communications systems, and so on) and for demonstrating new mission concepts.
- **As payloads outside of Earth orbit**
  To provide additional measurements or support for mothership spacecraft or to place instruments in advantageous orbits that would otherwise require dedicated launches.

- **The realization of megaconstellations at achievable prices**
  The use of universal design specifications and evolved industry standards have encouraged CubeSat components and sometimes entire spacecraft to be easier and cheaper to mass produce, significantly reducing costs in producing many for a constellation. The more spacecraft in a constellation, the more coverage is guaranteed, the greater redundancy is built-in, and the shorter revisit time is available.

3 CubeSats enhancing traditional satellite missions and objectives

This section will analyze how the CubeSats have enhanced the concept of traditional satellite missions related to Earth observation, telecommunications, and astronomy.

3.1 Earth remote sensing

Earth observation (EO) involves the use of either active or passive sensors to collect data related to a variety of different targets on the Earth below, ranging from (quite literally) the depths of the ocean to the highest mountain and beyond to the atmosphere and magnetosphere, as well as the mantle and core. Typically a CubeSat's small size limits the structure available for solar arrays to be mounted, and hence, its power output is limited. This means the majority of EO CubeSats have relied on passive sensors that collect data using the Sun or the Earth itself as the source and observing changes in the spectra received as it interacts with various parts of the atmosphere or surface [1]. As will be discussed later in Section 7, CubeSats' affordability and ease of mass production encourage their use in constellations, which greatly improves the quality of data returned for EO missions.

ExoCube is one example of the CubeSat specification being used as a platform to support dedicated Earth observation science. This 3U launched in January 2015 measured the density of hydrogen, helium, nitrogen, and oxygen in the Earth's exosphere, in addition to characterizing the ion density above various ground stations [6]. As a CubeSat, it was capable of supporting all the instruments required for this and proved that such miniaturization is feasible for a CubeSat-sized platform. Other examples of EO missions are reported in Table 1.

3.2 Telecommunications

CubeSats have taken a prominent role in demonstrating in orbit technologies for telecommunications since their earliest years. Arguably out of necessity, increasingly miniaturized communication technologies are demanded for CubeSats, as it is often the downlink bandwidth that limits the science data return rather than the payload instrument's capability—which typically generates more data than can be downlinked from a CubeSat [1]. CubeSats' affordability and commonplace use of industry standards ensure that any new developments tested on board have a more streamlined

Table 1 Examples of Earth observation CubeSat missions.

Spacecraft, size, and launch	Organizations involved	Mission description
ExoCube, 3U, 2015	California Polytechnic State University [6]	Uses an Ion and Neutral Mass Spectrometer (INMS) to characterize the densities of various elements and ions in the Earth's exosphere and ionosphere
ANGELS, 12U, 2019	CNES and Hemeria [7]	Carries the newest version of "Argos" tracking payloads, which currently are on a constellation that locates and tracks beacons tagged onto animals, ships, and weather buoys
RAVAN, 3U, 2016	NASA [1]	Measures reflected solar and emitted thermal energy from the Earth to more accurately determine radiative forcing
RaInCube, 2018	NASA JPL [1]	A technology demonstration mission to enable Ka-band precipitation radar technologies on a low-cost, quick-turnaround platform (see also Table 2)
SathyabamaSat, 2U, 2016	Sathyabama University [1]	Carries an infrared spectrometer for measuring levels of greenhouse gases and pollutants in the atmosphere

route to becoming marketable components in their own right. Further to this, CubeSats' regular form, components, and design encourage their ability to be mass produced for constellations that are vital for ensuring near-constant coverage of target ground areas. Representative examples of CubeSats that have demonstrated advanced telecommunications technologies are listed in Table 2.

3.3 Astronomy

Most astronomical missions require specific objectives and dedicated instruments that provide impeccably precise, restricted data about unique astronomical bodies—the Sun, planets and moons, distant galaxies, black holes, and exoplanets, to name a few. It is often impossible to construct payloads that can be applicable to multiple types of targets; hence, often dedicated satellites are required for each payload or suite of instruments if the customer can afford them. More often a space agency is responsible for operating a larger satellite with a suite of instruments focused on one target type and distributing the data to relevant customers afterward (GAIA, Hubble, SWIFT, Spitzer, etc.). CubeSats however present the opportunity for individual or a consortium of customers to develop a dedicated spacecraft, capable of supporting their unique requirements for specific targets at much lower costs than with previous, larger, standard buses [2].

One such experimental CubeSat dedicated to astronomy was ASTERIA, launched in 2017 by JPL and MIT. ASTERIA was a 6U, exoplanet hunter that operated using

Table 2 Examples of telecommunications technology related CubeSat missions.

Spacecraft, size, and launch	Organizations involved	Mission description
ISARA, 3U, 2017	NASA/JPL [1]	Proof of concept for a foldable Ka-band reflectarray antenna (integrated into the solar arrays) that would significantly improve downlink bandwidth for small satellites
CQuCoM, 6U, undetermined	University of Strathclyde/ National University of Singapore [8]	An IoD of quantum key distribution for future communication and as a pathfinder for potential future constellations
RaInCube, 6U, 2018	NASA/JPL [1]	Uses the first Ka-band radar from a CubeSat (including deployable antenna dish) that was used to track storm movements
CubeSOTA, 6U, undetermined	NICT/University of Tokyo [9]	This LEO CubeSat seeks to use a GEO satellite as a relay to benefit from its much-improved coverage and link availability while sacrificing some bandwidth
CubeRRT, 6U, 2018	NASA/JPL [10]	IoD of in situ radio frequency interference filtering, which will improve the accuracy of reading microwave radiances from geophysical sources such as soil moisture and sea-surface salinity

the transit method to observe stars several times over a period of a few days and measuring any dips in the light curve due to an exoplanet occulting its star. This impressive accomplishment is testament to the CubeSat community's professionalism, due to its use of COTS components to achieve the accurate pointing control required. Not only did the CubeSat operate successfully for over 2 years (with an original mission lifetime of 90 days), but also it was the first CubeSat to observe a transiting ExoPlanet—55 Cancri e [11,12]. Other notable examples of CubeSats that successfully performed astronomy and astrophysics research are shown in Table 3.

4 CubeSats supporting space-borne experiments

The majority of in situ microgravity research conducted today happens aboard the International Space Station (ISS). This platform acts not only as a testament to international collaboration but also as the most sophisticated and well-equipped laboratory in Earth orbit. Outside of the ISS a selection of other platforms and satellites regularly

Table 3 Examples of CubeSat missions with astronomy science objectives.

Spacecraft, size, and launch	Organizations involved	Mission description
ASTERIA, 6U, 2017	MIT/NASA/JPL [2,11]	First successful exoplanet detecting CubeSat equipped with COTS systems for providing necessary pointing accuracy
HaloSat, 6U, 2018	NASA/GSFC/JHU/Nagoya University of Japan [1,2,12]	Measures x-ray emissions from highly ionized oxygen in the Milky Way's galactic halo. Galactic halos are theorized to be a possible reservoir of baryons currently "missing" from the observed local universe, compared with cosmic background microwave radiation
MinXSS, 3U, 2015	University of Colorado Boulder [2]	Characterize the solar x-ray spectrum across wavelengths where the largest enhancement of solar flares is expected to occur
CSSWE, 3U, 2012	University of Colorado Boulder [3]	Measures the relativistic energy spectrum of solar energetic particles for studying space weather. Also features a magnetometer. Supports other missions such as VA probes
PicSat, 3U, 2018	Paris Observatory, PSL, CNRS [13]	A dedicated CubeSat for the study of the beta Pictoris system, its circumstellar disk, exocomets, and the transit of the Hill Sphere of exoplanet beta Pictoris b

conduct or have conducted crucial research that can only be achieved in the environmental conditions of space and microgravity. The fields of research that benefit from research experiments in space are numerous and include materials science, crystallography, planetary science, molecular chemistry, fiber optics, pharmacology, and biology.

In addition to the ISS, several of the larger manned platforms include MIR, Tiangong 1 and 2, and STS. However, using humans to operate or install (or even share an enclosed environment with) experiments while in space is a very costly and complex task—but one that has a great many advantages too. At times, it may be more cost-efficient to have an automated system on board a satellite or space station that can be teleoperated from the ground instead; however, the cost of developing a satellite that can simultaneously prepare, operate, monitor, and transmit the results of as wide a variety of experiments that exists or are desired for spaceflight is nearly impossible. Often, these satellites carry suites of payloads involving intersecting areas of science or experiment requirements—for example, Bion-M1 carried a selection of animals that required roughly the same ambient environmental requirements [14].

CubeSats however can be tailor-made to suit specific and unique payloads desired by research groups—often with lower costs and development time than one may need to fly similar payloads on board the ISS or other platforms, manned or otherwise. Although not as fully developed as CubeSat payloads for Earth observation or astronomy, exploiting this niche and market demand will greatly expand the variety and depth of science conducted in space today [2]. These opportunities not only allow long-time space science research groups the chance to have near-full control of their payload and its conditions but also provide industries with no previous relations to space an open door to affordably test any materials or drugs or procedures in space [15].

One field that is likely to see future growth from adopting CubeSats is the study of life sciences in space. While vitally important for preparing for longer-term space missions, both beyond Earth to the Moon or Mars, and further studying the effects of spaceflight in Earth orbit, this research will doubtless be used to benefit surface-dwelling people on Earth too [16,17]. Studies of cancer, pathogens, and age-related diseases and new methods of producing medicine and treatments for these already are an important highlight of ISS science operations and a field that is planned to mature during the commercialization of the ISS, expansion of private spaceflight, and preparation for deep space exploration.

Currently, NASA Ames' Small Satellites Technology Division has been the premier source of CubeSats with biological payloads—although notably SpacePharma launched DIDO 2, and there have been other larger biosatellites that have flown as well [1]. There is even a 6U named BioSentinel on the manifest for interplanetary space on Artemis I. In the future, life sciences in space could be expanded from the ISS and into the CubeSat community in the same way that EO and astronomy are being expanded today. If simple, affordable CubeSat platforms were to be developed to support the wide variety of payloads and instruments that are wanted to be flown, it would revolutionize the way in which science is taken from Earth labs to orbit. Using the CubeSat standard as a template model to support space qualified payloads originally modified from their control experiments in the lab would greatly streamline the process of conducting spaceflight biology research. Table 4 lists representative examples of CubeSats that have flown dedicated space environment experiments.

5 CubeSats as technology demonstrators

CubeSats make excellent platforms for technology demonstrations and for proof-of-concept missions. Multiple factors support this, such as their widely accepted industry standards, design specifications, and COTS equipment providing simple and cheap spacecraft buses and structures that can be relatively cleanly built around a more complex, dedicated payload. This "standard packing" is also a useful factor for encouraging streamlined interfacing between a complex and unique satellite and the launch provider/deployment mechanism. The other prime advantages are that of the low cost and buy-in for a CubeSat mission is more practical for early-stage businesses and researchers with less capital to invest in a technology with no flight heritage and also

Table 4 Examples of CubeSats supporting space-laboratory payloads.

Spacecraft, size, and launch	Organizations involved	Mission description
PharmaSat, 3U, 2009	NASA/AMES [2,16]	Grew “brewer’s yeast” in LEO before dosing it with various concentrations of an antifungal agent to determine its efficacy in orbital conditions
BioSentinel, 6U, Artemis I	NASA/AMES [18]	A secondary payload on Artemis I flying heritage from PharmaSat to test “brewer’s yeast” growth and DNA repair in the radiation environment beyond Earth orbit
Q-PACE, 3U, slated 2020	University of Central Florida [2]	Studies the early stages of protoplanet accretion by observing collision of small, 0.1 mm–1 cm, particle collisions inside the CubeSat’s payload chamber
3Cat-1, 1U, 2018	Universidad Politécnica de Cataluña [19]	Contains seven payloads in a 1U structure, one to categorize the performance of graphene field-effect transistor

for nonspace industries to work with specialized consultants to develop their ideas into flyable payloads in a cost-effective and streamlined manner.

One important quality that may be overlooked regarding CubeSats is that as a relatively, inexpensive platform for space operations, risky in orbit demonstration missions can be seen as more economically viable, as less money and technology is “lost” should the mission fail. Many of these technology demonstrator missions would pose risks deemed uneconomical or downright dangerous had they been carried out aboard larger research satellites or the ISS; however, given the comparatively low investment required for a CubeSat mission, some technology can be demonstrated efficiently and effectively. Notable examples of technology demonstration missions are provided in Table 5.

An example of how CubeSats that can support in-orbit demonstration of new technologies has been a part of the RemoveDEBRIS mission from SSTL and the University of Surrey. During this mission, a mothership measuring $65 \times 65 \times 72$ cm and two 2U+ CubeSats were deployed together from the ISS (incidentally the largest deployment from the ISS so far), with the first CubeSat, DebrisSat1, deploying an inflatable aluminum balloon to be the target of a net released from the mothership. DebrisSat 2 contained instruments that were used to verify the precision of the mothership’s vision-based navigation experiment and other tests for the validation of the technology [20].

LightSail 1 and 2 were 3U CubeSats designed to test the deployment and operation of solar sailing in LEO. Solar sails had been attempted previously, with the most successful certainly being the JAXA IKAROS mission that used controlled solar sailing to reach and fly past Venus [23].

Table 5 Examples of CubeSat technology demonstrator missions.

Spacecraft, size, and launch	Organizations involved	Mission description
RemoveDEBRIS, Mothership +2 × 2U, 2018	SSTL/University of Surrey/other industrial partners [20]	Designed to demonstrate four different methods of active space debris removal, including nets, harpoons, drag sails and vision-based navigation for an autonomous satellite. One Cubesat was the target for the harpoon and the other a “target” for the mothership to maneuver around and collect data
STARS flights, 2U, various launches	Kagawa University [21]	Demonstrations of space elevator and tether technology. Each “U” of CubeSat separates and unspools a tether between them to test the movement along it of a “cable car”. While full-size space elevator technology is in its extreme infancy, some examples of it are already now TRL 9 thanks to CubeSats
LightSail 2, 3U, 2019	The Planetary Society [22]	Tested controlled solar sailing in LEO on board a CubeSat. Results are debated although IKAROS achieved solar sailing to Venus in 2010
DIDO-2, 3U, 2017	SpacePharma [15]	Designed to test various biochemical processes in space that are vital for further understanding life sciences in space

6 CubeSats as deep space explorers

CubeSat missions are not limited to low Earth orbit (LEO) but can be considered as a tool to improve knowledge of deep space or as support for future manned missions to other planets. In this section, recent missions that have demonstrated the capability of CubeSats to operate beyond Earth orbit will be analyzed: from first successes with the MarCO mission to upcoming launches to the Moon and NEOs that CubeSats will rideshare.

Several critical technology gaps have had to be crossed for successful use of CubeSats in deep space operations. For one the small structure of CubeSat restricts the aperture size of antenna that can be used—similar to solar array size and imaging apertures for astronomical instruments—which presents an issue in possessing a large enough effective area for sufficiently high transmission or reception. Several solutions to this issue have been flown in Earth orbit, including patch and circularly polarized antennas and reflectarrays [18]. Another issue is related to the capabilities that miniaturized propulsion devices can provide, both in terms of total delta-v and thrust, which can be problematic for CubeSats reaching and establishing orbit at their target destination within a reasonable and cost-effective time frame. Fortunately, however, the ability of CubeSats to be manifested as secondary payloads onto launchers through standardized deployers means that they take advantage of rideshare opportunities on interplanetary missions that larger secondary spacecraft might not be able to afford. With the upcoming renaissance of interplanetary exploration, supported by a renewed space race and new super heavy lift vehicles, it may be more likely that we see CubeSats flying alongside their motherships into the solar system.

As the key driving factors and other sections of this chapter describe, CubeSats have found their place supporting a wide variety of roles and missions across the space sector. The utility of the platform and success of the supporting industry have ensured that developing new prototypes and flight ready pathfinding missions is simple, cost-effective, and easy to facilitate for all involved parties. These benefits encourage mission planners and PIs looking into undertaking deep space missions on a budget (or with a desire to demonstrate relevant technology) to utilize the CubeSat as the desired platform. We have already explored how CubeSats can support miniaturized instruments with a variety of functions—EO, astronomy, telecommunications, etc., so it is unsurprising that these techniques and advantages are desired to be leveraged for interplanetary exploration also. Not only are the support for instrumentation available for interplanetary CubeSat developers one appealing factor, but also is the support available—in terms of COTS products, experience, and facilities available—for other subsystems, launch support, mission analysis, and rideshare opportunities.

A list of the main CubeSat missions beyond Earth orbit is reported in Table 6.

Table 6 Examples of CubeSat missions beyond Earth orbit.

Spacecraft, size, and launch	**Organizations involved**	**Mission description**
MarCO, 2 × 6U, 2018	NASA/JPL [1,18]	The first interplanetary CubeSat mission. Daughter spacecraft to the InSight lander, these relayed critical EDL information to Earth in near real time when the lander or other orbiters such as MRO were unable to (see Chapter 4)
Lunar Exploration from Artemis I, 13 × 6U, NEL 2021	Various/NASA Launch [1,18]	The first launch of SLS to cis-lunar space will allow for these CubeSats to travel beyond Earth orbit and into deep space. Many are focused on studying the Moon
AIDA, 3 CubeSats from two motherships, 2021 and 2024	NASA/ESA [12,24]	Daughter CubeSats to assist with studying a near-Earth object (65,803 Didymos currently) and to analyze the effectiveness of intentional impact by a spacecraft in redirecting its orbit path
NEA Scout	NASA [18]	Solar sail propelled spacecraft from Artemis I. Planned to sail to a near Earth asteroid and perform analysis on flyby.

7 CubeSats as distributed instruments in constellations

CubeSats have been implemented to fulfill GNSS and EO roles as individual satellites and constellations. However, in the future, their ability to be mass produced inexpensively and mass deployed safely has led to their being featured prominently in plans for future megaconstellations such as Planet Labs and Spire Global, as well as other upcoming constellations like Kepler Communications. Owing to their cost advantage, more spacecraft can be placed in a constellation for the same price; therefore more coverage is guaranteed, greater redundancy is built-in, and shorter revisit times are made available [1,25].

These advantages become even more crucial and provide more value for cost when they are used in scenarios such as disaster management where up-to-date, high-quality observation data can be vital for directing emergency services, warning populations of imminent danger, and reliably providing satellite-internet communications to workers on the ground in the event of damaged telecom infrastructure. It is for these reasons that emerging space powers, NGOs, and developing countries are increasingly seeking to develop or utilize CubeSat constellations for cost-effective disaster management resources [25]. Table 7 lists representative examples of CubeSat-based megaconstellations.

Table 7 Examples of CubeSat-based megaconstellations.

Spacecraft, size, and launch	**Organizations involved**	**Mission description**
Planet Labs	Planet Labs [1]	A series of "flocks" of various Cube and SmallSats equipped with telescopes to image in high-resolution swathes of Earth within 52 degrees of the equator—a large portion of Earth's agriculture and population
Spire Global	Spire Global [1]	Over 80 operational CubeSats in a constellation offering space as a service. It measures the atmospheric refraction of radio signals sent from GNSS satellites on the other side of the planet
Kepler Communications	Kepler Communications [26]	An upcoming company planning on using a constellation of CubeSats to provide high-bandwidth links to relay satellites and ground stations, including for polar operations

8 Conclusions

The variety of missions and applications that CubeSats can undertake—and very often at significant cost reductions—has led to them being widely regarded as the economical method of operating in space. While initially designed to be a tool for university students to gain experience with real space hardware, they have seen adoption from space agencies, military organizations, nonprofits, and from some of the biggest aerospace industries on Earth today. The CubeSat community has grown into an important sector of the space economy and is responsible for driving the ambition, accessibility, and variety of missions conducted today, thanks to the evolution of common industry standards, COTS products and services for developers, and mass production. The adoption of CubeSats by emerging space powers, NewSpace ventures, professional space agencies, and traditional aerospace industries reaffirms their position as viable and practical platforms for spaceflight and assures their continuing utilization in the future. The future of CubeSat applications largely depends on the creativity and capability of the CubeSat developers to identify and solve interesting new problems using small, innovative satellite solutions.

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